

TECHNICAL REPORT: Rescue Mission Optimization

AI Project: Comparative Study of Search Algorithms vs. Genetic Algorithms

Team Information

- **Project Title:** Rescue Mission Order (TSP Variant)
 - **Team Members:** Ibrahim Hashem, Ahmed Salah, Amany Mahmoud
 - **Programming Language:** Python 3
 - **Core Libraries:** NumPy, Matplotlib, Random, Collections (deque)
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1. Environment Explanation

The environment is a 20×20 2D Grid.

Obstacles

Complex wall patterns are placed vertically and horizontally to block direct paths, making the problem non-trivial and challenging.

Entities

- **Robot (R):** Starting at position (1, 1)
- **Victims (V1–V5):** Five victims placed in strategic corners of the grid to force the robot to traverse the entire map.

Logic

The grid uses a binary representation where:

- 1 represents a wall
 - 0 represents free space
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2. Search Algorithm: BFS vs. DFS

What Search Algorithm is Used?

We use Breadth-First Search (BFS) to calculate the true distance between all points.

Why BFS? BFS is chosen because it guarantees finding the shortest path in an unweighted grid.

The Problem with DFS

Depth-First Search (DFS) focuses on depth and might find a very long and winding path before reaching the target. For a rescue mission, we need the minimum number of steps to calculate an accurate Fitness Matrix.

Output

The BFS generates a 6×6 Distance Matrix representing the actual distances between the Robot and all five victims.

3. Genetic Algorithm (GA) Components

This section explains the core logic of the Genetic Algorithm.

3.1 Fitness Function: What and Why?

Definition

The Fitness Function calculates the Total Distance of a specific rescue sequence.

Formula

$$\text{Fitness} = \text{Distance}(\text{Robot} \rightarrow \text{First Victim}) + \sum \text{Distance}(V_i \rightarrow V_{i+1})$$

Why?

This models the real-world objective: minimizing the travel time required to save all victims.

A lower fitness value represents a better solution.

3.2 Selection Algorithm: Tournament Selection

Type

Tournament Selection with tournament size = 5.

Why?

It balances:

- Selection Pressure (favoring stronger solutions)
- Diversity (allowing weaker but unique solutions to survive)

This helps prevent the algorithm from getting trapped in a local optimum too early.

3.3 Crossover: Ordered Crossover (OX)

Type

Ordered Crossover (OX)

Why?

Because our chromosome represents a Permutation Problem.

Traditional crossover methods (such as One-Point Crossover) may produce invalid solutions containing duplicate victims, for example visiting V2 twice.

Ordered Crossover preserves the relative order while ensuring every victim is visited exactly once.

3.4 Mutation: Swap Mutation

Type

Swap Mutation

Why?

By swapping two random victims in the sequence, we introduce Genetic Diversity.

This random change allows the algorithm to explore new areas of the solution space that crossover alone might not discover.

3.5 Termination Phase: When Does It Stop?

Condition

A fixed number of 500 generations.

Reasoning

Analysis shows that for five victims, the optimal solution (68 steps) is usually found within the first 100 generations.

Running for 500 generations provides a safety buffer to ensure the solution is stable and fully optimized.

4. Visualization & Benefits

Explanation

The visualization displays:

- **The grid**
- **Black walls**
- **The robot (green square)**
- **Victims (colored circles)**

Arrows

Arrows represent the rescue sequence found by the Genetic Algorithm.

Benefit

The visualization provides Immediate Validation.

It proves that:

- The path is physically possible
 - All walls are avoided
 - The sequence follows the shortest-path logic determined by BFS
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5. Comparison: Greedy Algorithm vs. Genetic Algorithm

Greedy Algorithm

The Greedy Algorithm always chooses the nearest victim next.

In simple environments, it is fast and often effective.

Genetic Algorithm Advantage

The Genetic Algorithm performs better in complex environments where selecting the nearest victim may lead to:

- Dead ends

- Longer total travel distance later

The GA evaluates the entire rescue sequence globally, not just the immediate next step.

Appendix: Manual Calculation (For Pen & Paper Defense)

If asked to explain the logic manually:

Step 1: Input

Example chromosome:

[4, 1, 5, 3, 2]

Step 2: Lookup

Use the Distance Matrix generated by BFS.

Step 3: Summation

- **Robot → V4 = 10 steps**
- **V4 → V1 = 14 steps**
- **V1 → V5 = 14 steps**
- **V5 → V3 = 16 steps**
- **V3 → V2 = 14 steps**

Step 4: Final Result

Total Distance = 68 steps